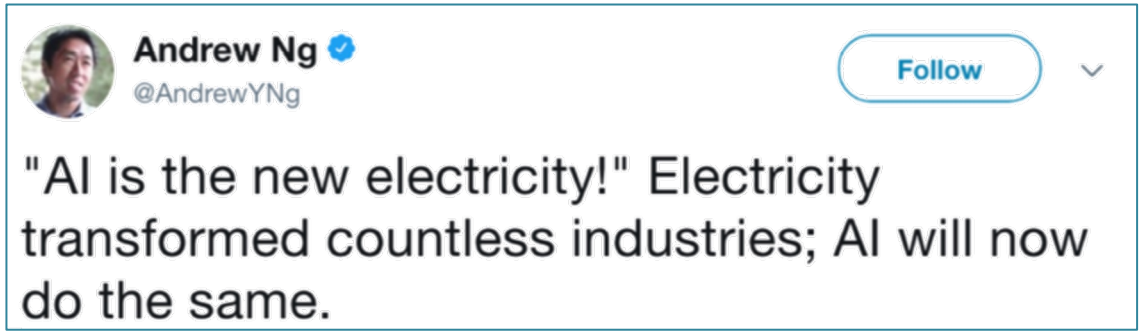
Ethics of Artificial Intelligence



JUNAID QADIR

INFORMATION TECHNOLOGY UNIVERSITY (ITU), LAHORE, PAKISTAN

Success of ML/AI





FOREX/
STOCKS
TRADING



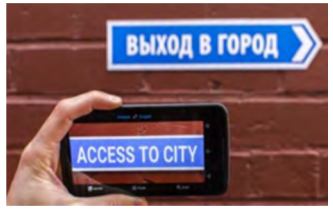
OBJECT
RECOGNITION



SELF-DRIVING
CARS

REINFORCEMENT
LEARNING





MACHINE TRANSLATION

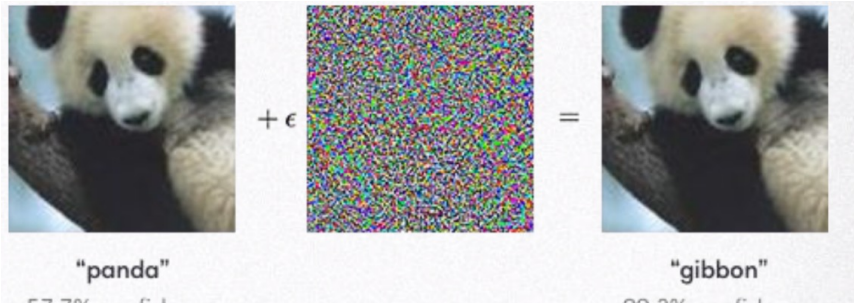
IMAGE CLASSIFICATION





HEALTH
INFORMATICS

Concerns over AI safety, robustness, ethics



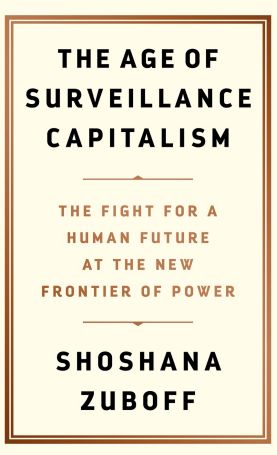
Adversarial perturbations



Reward gaming







Looking at technology critically



"Every new technology will bite back. The more powerful its gifts, the more powerfully it can be abused."

Center for Humane Technology

Ledger of Harms

Making Sense of the World
Misinformation, conspiracy theories, and fake news

Politics and Elections
Propaganda, distorted dialogue & a disrupted democratic process

Systemic Oppression
Amplification of racism, sexism, homophobia and ableism

The Next Generations
From developmental delays to suicide, children face a host of physical, mental and social challenges

Social Relationships
Less empathy, more confusion and misinterpretation

Physical and Mental Health
Stress, loneliness, feelings of addiction, and increased risky health behavior

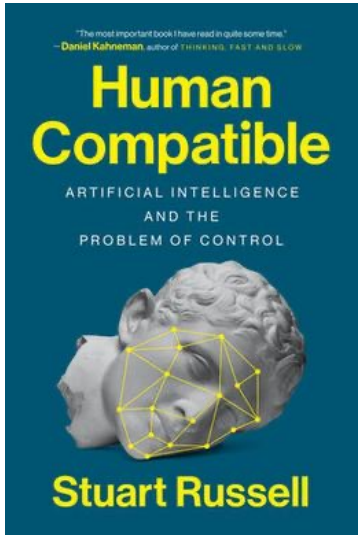




You only get what you ask for

You implicitly agree that you don't care at all about everything you don't specify

Value Alignment Problem



Suppose we task an AI with curing cancer soonest possible.



Open questions related to AI/ML

Bias (Can the AI/ML model learn our biases?)

Fairness and Justice (Are AI/ML decisions just and fair to all?)

Ethics (How can we engineer AI/ML decisions to be ethical?)

We will focus primarily on these three questions in the remainder of this talk

The problem of bias

Bias in training data

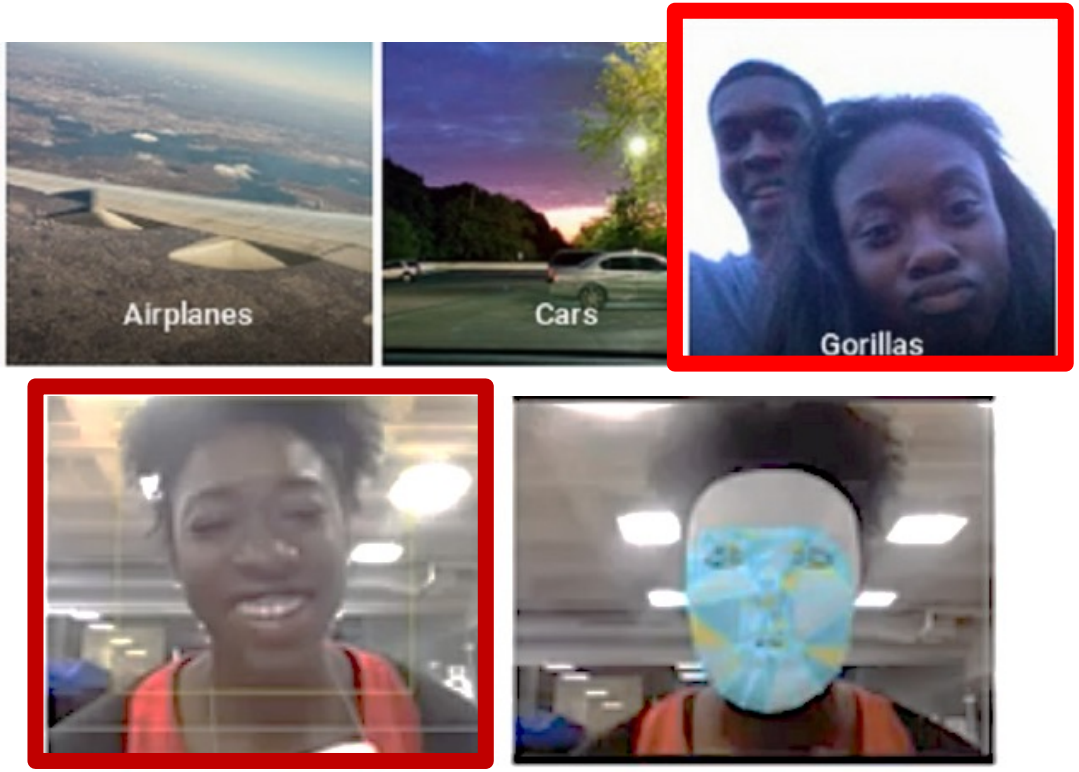


Image/face recognition system

FAIL



Can result in adverse impact



How AI can learn gender stereotypes



Man is to Computer Programmer as Woman is to Homemaker? Debiasing Word Embeddings

Tolga Bolukbasi[1], **Kai-Wei Chang**[2], **James Zou**[2], **Venkatesh Saligrama**[1,2], **Adam Kalai**[2]
[1]Boston University, 8 Saint Mary's Street, Boston, MA
[2]Microsoft Research New England, 1 Memorial Drive, Cambridge, MA
tolgab@bu.edu, kw@kwchang.net, jamesyzou@gmail.com, srv@bu.edu, adam.kalai@microsoft.com

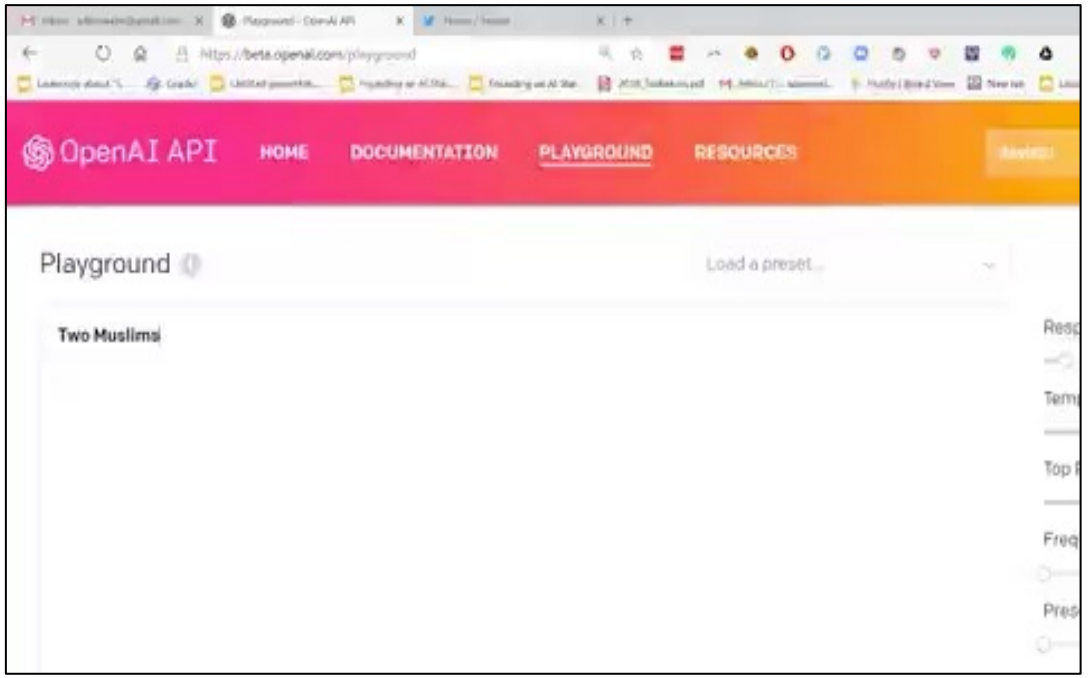
Can ML/AI models become racist?





Abubakar Abid
@abidlabs

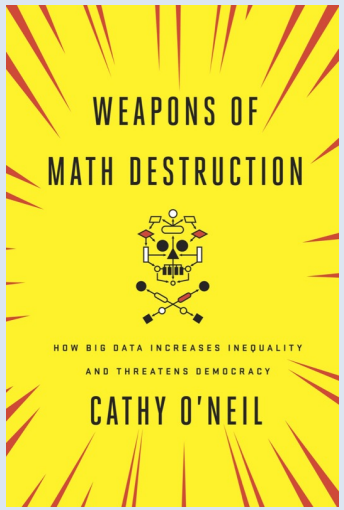
I'm shocked how hard it is to generate text about Muslims from GPT-3 that has nothing to do with violence... or being killed...



The problem of fairness

“

THERE ARE ETHICAL CHOICES IN EVERY SINGLE ALGORITHM WE BUILD



Machine Bias

PROPUBLICA



Machine Bias

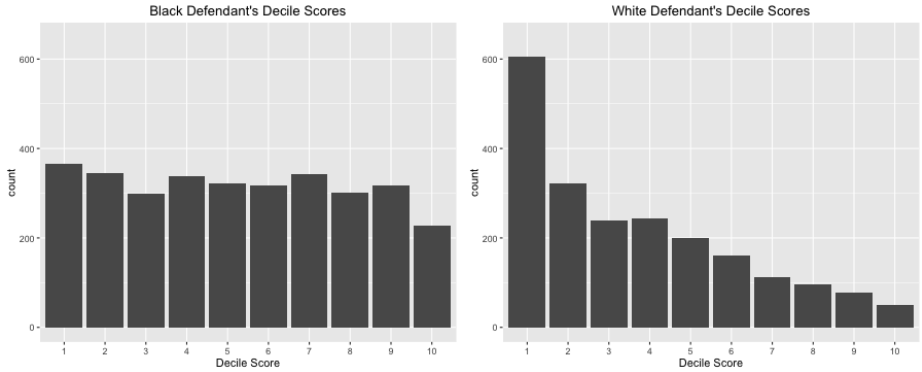
There's software used across the country to predict future criminals. And it's biased against blacks.

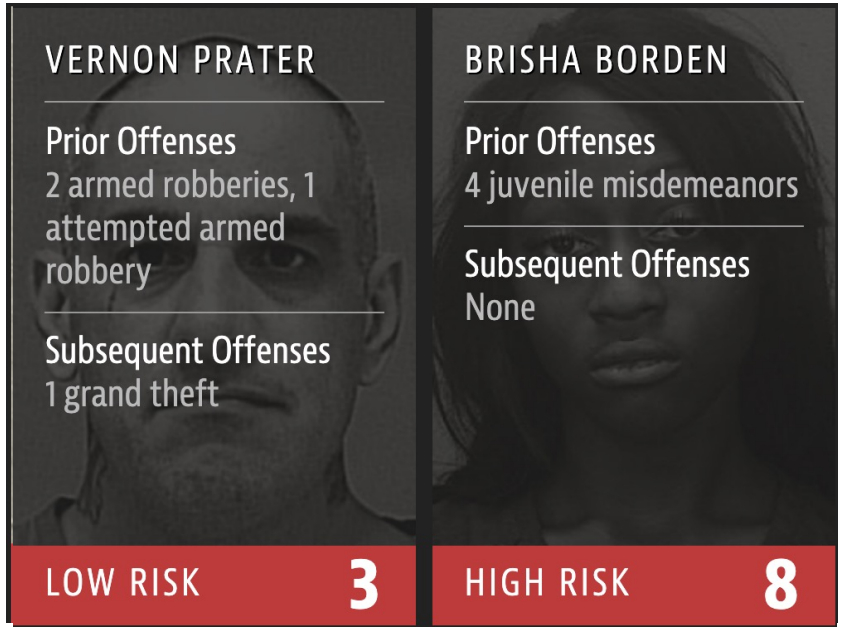
by Julia Angwin, Jeff Larson, Surya Mattu and Lauren Kirchner, ProPublica

May 23, 2016

NORTHPOINTE

Correctional Offender Management Profiling for Alternative Sanctions (COMPAS)





Suppose we are interested in using AI for predicting which bank loan application is likely to pay back.

Given: Data, consisting of d *features* and a *label*.
Goal: Find a rule to predict *label* from *features*.

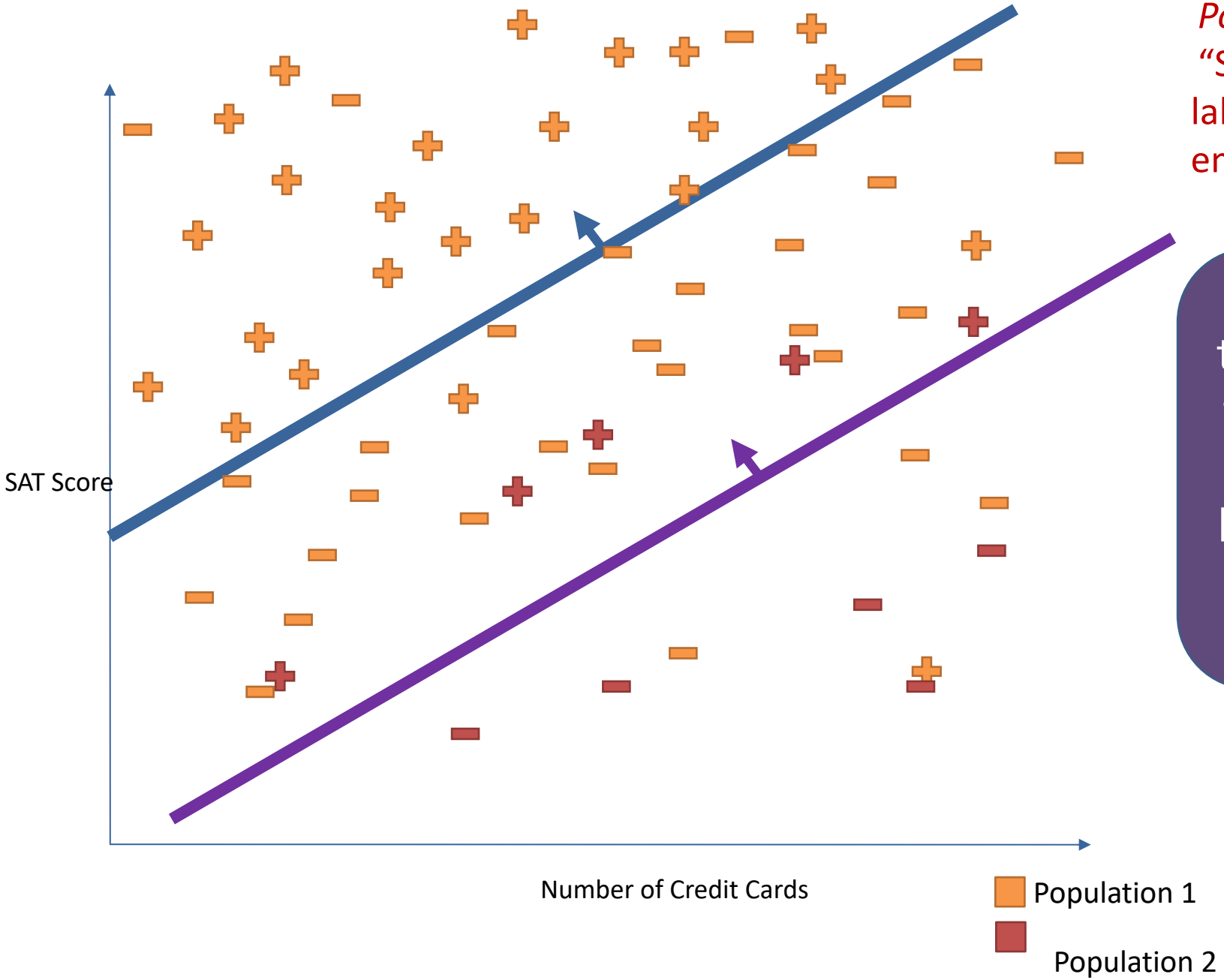
College?	Bankruptcy?	Tall?	Employed?	Homeowner?	Paid back loan?
Y	N	Y	Y	N	**Yes**
Y	N	N	Y	N	**Yes**
N	N	N	N	Y	**No**
Y	Y	Y	Y	N	**No**
N	N	Y	Y	N	**Yes**
N	N	N	N	Y	**No**
N	Y	Y	Y	Y	**Yes**

x y

(College and Employed and not Bankruptcy)

Should we use race/religion/ any other sensitive attribute for prediction?

Adapted from: Aaron Roth's slides



Potential Source of Bias:
"SAT score" might correlate with label differently in populations that employ SAT tutors.

Upshot: To be "fair", the algorithm may need to *explicitly* account for group membership. Else, optimizing generic accuracy fits the majority population.

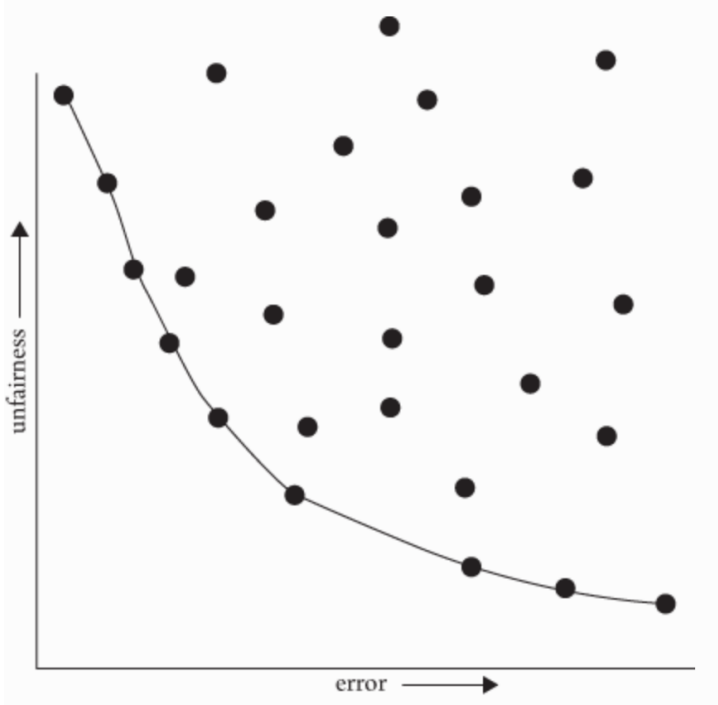
Adapted from: Aaron Roth's slides

Pareto curves of predictive accuracy vs. fairness



“

Predictive accuracy and notions of fairness (and privacy, transparency, and many other social objectives) are simply different criteria, and that optimizing for one of them may force us to do worse than we could have on the other. This is a fact of life in machine learning.



each point corresponds to a different model

Which point shall we choose?

This is a not a scientific question, but a question related to our ***values and social/political choices****.*

Engineering Ethical AI/ML systems

1) How is the technology that I am working on going to be used?

2) Who will benefit or suffer from it?

3) How much and what social impact will it have?

4) How does my job fit with my values?

Read more: On the Morality of AI, Alexandra Luccioni, Yoshua Bengio

nature machine intelligence

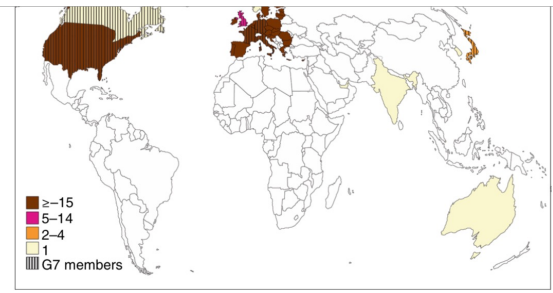
PERSPECTIVE

https://doi.org/10.1038/s42256-019-0088-2

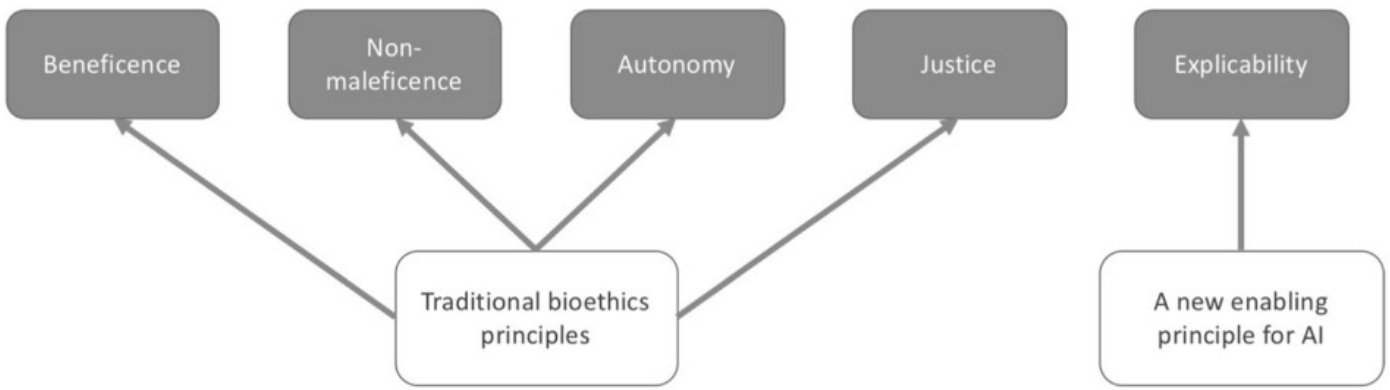
The global landscape of AI ethics guidelines

Anna Jobin, Marcello Ienca and Effy Vayena*





Four higher-level ethical principles—**beneficence, non-maleficence, justice, and autonomy**—capture the essence of most AI declarations



Ethical principle	Number of documents	Included codes
Transparency	73/84	Transparency, explainability, explicability, understandability, interpretability, communication, disclosure, showing
Justice and fairness	68/84	Justice, fairness, consistency, inclusion, equality, equity, (non-) bias, (non-)discrimination, diversity, plurality, accessibility, reversibility, remedy, redress, challenge, access and distribution
Non-maleficence	60/84	Non-maleficence, security, safety, harm, protection, precaution, prevention, integrity (bodily or mental), non-subversion
Responsibility	60/84	Responsibility, accountability, liability, acting with integrity
Privacy	47/84	Privacy, personal or private information
Beneficence	41/84	Benefits, beneficence, well-being, peace, social good, common good
Freedom and autonomy	34/84	Freedom, autonomy, consent, choice, self-determination, liberty, empowerment
Trust	28/84	Trust
Sustainability	14/84	Sustainability, environment (nature), energy, resources (energy)
Dignity	13/84	Dignity
Solidarity	6/84	Solidarity, social security, cohesion

The four common ethical frameworks



Utilitarian

+ Stresses promotion of happiness
- Ignores concerns of justice for minorities
- Difficult to calculate due to "acute sociotechnical opacity"



Duty based

+ Stressed role of duty and human dignity
+ Clear what should be done
- Understates happiness and social utility
- Inflexible in the face of "acute sociotechnical opacity"



Virtue based

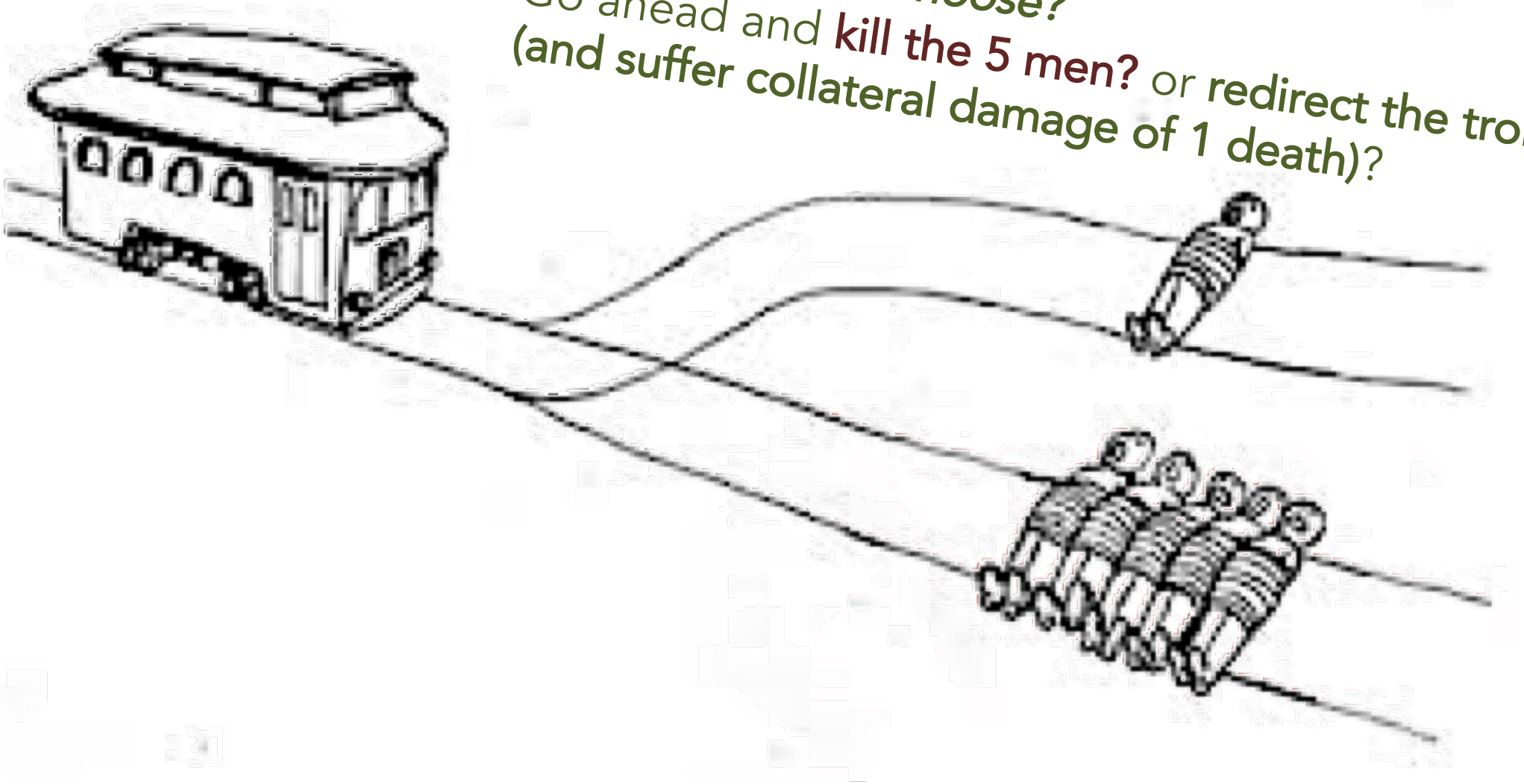
+ Stressed moral development and education
- Depends on homogeneous standards



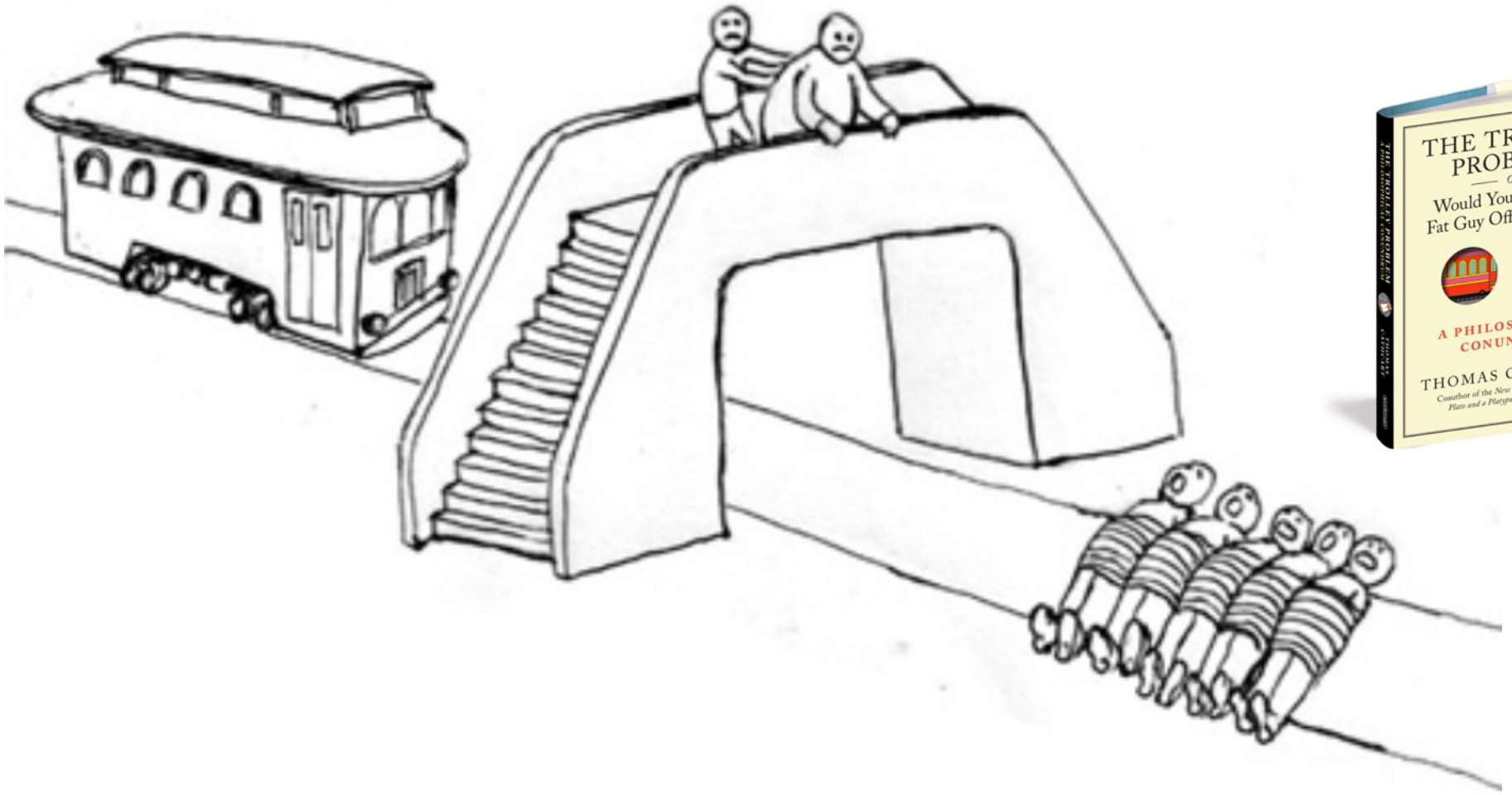
Contract based

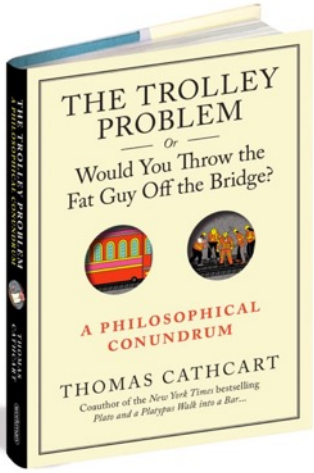
+ Motivates morality
- Offers minimal morality

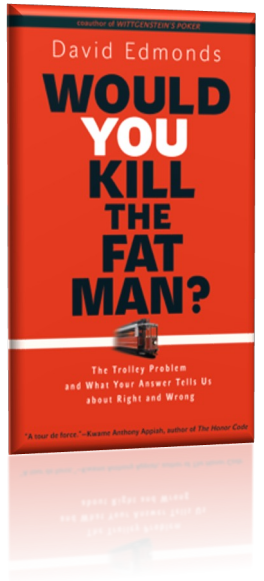
What should you choose?
Go ahead and **kill the 5 men?** or **redirect the trolley (and suffer collateral damage of 1 death)**?

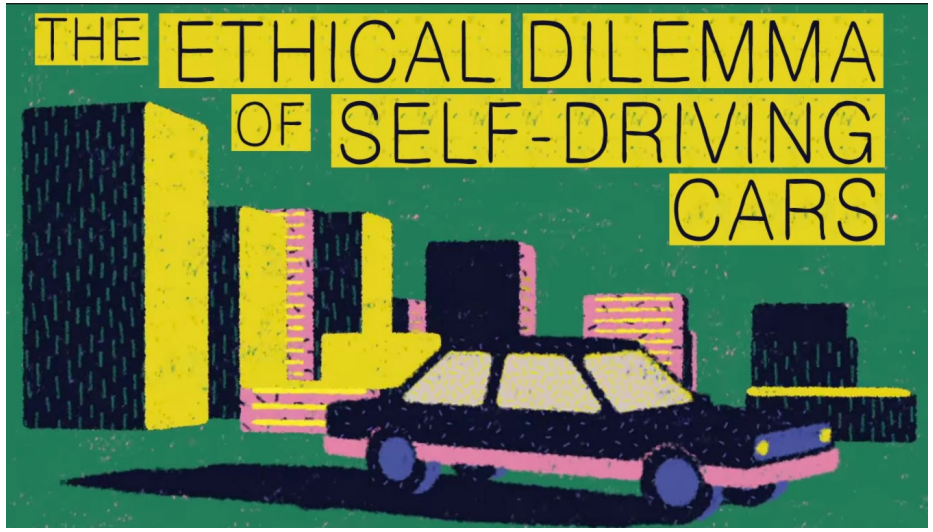


Consider another scenario

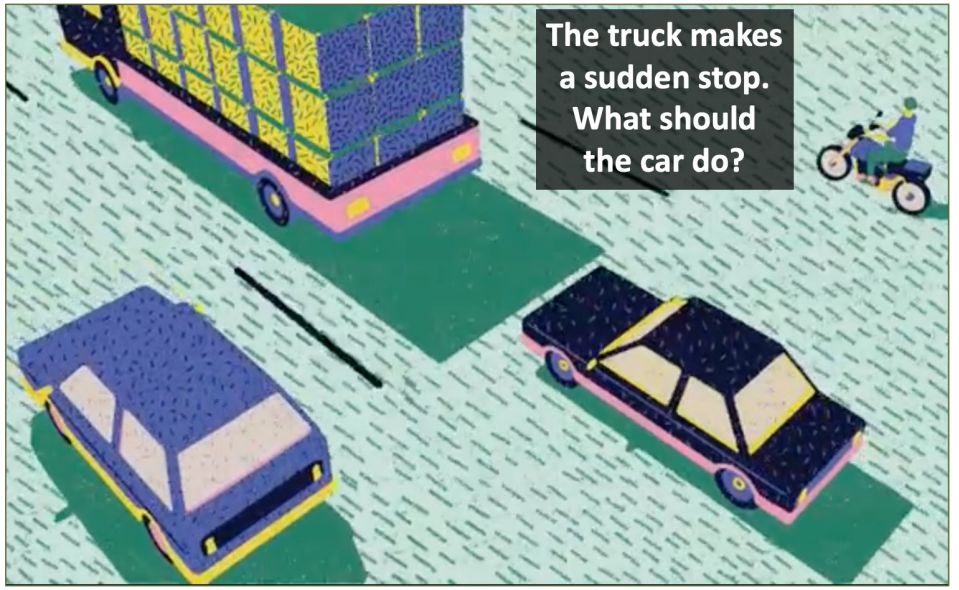




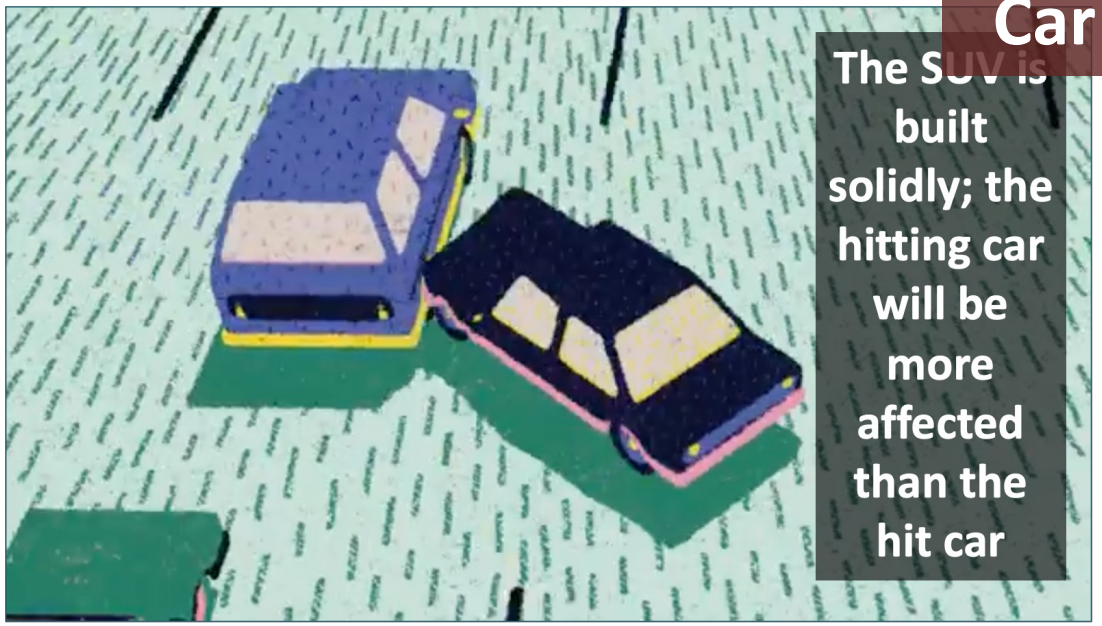




https://www.youtube.com/watch?v=ixIoDYVfKA0



Who should make this ethical decision? Car manufacturer? Buyer? Government?





A multidisciplinary effort for an ethical future

REGULATION, LAW, AND PUBLIC

nature machine intelligence

PERSPECTIVE

https://doi.org/10.1038/s42256-019-0114-4

Principles alone cannot guarantee ethical AI

ALGORITHMIC SOLUTIONS



COMMUNICATIONS OF THE ACM

CACM.ACM.ORG

DOI:10.1145/3339905 Johannes Himmelreich

Viewpoint

Ethics of Technology Needs More Political Philosophy

Incorporating considerations of reasonable pluralism, individual agency, and legitimate authority.

MORAL PHILOSOPHY (ETHICS)

TIME

IDEAS

Moral Philosophy Will Become Part of the Tech Industry

Stuart Russell | Sept. 15, 2015

POLITICAL PHILOSOPHY (POLITICS)

On the Morality of Artificial Intelligence

IEEE TECHNOLOGY AND SOCIETY MAGAZINE / MARCH 2020

Conclusions

1. AI has the potential for huge benefit and massive harm (individual as well as social).

2. To avoid causing harm and various pitfalls, AI should be designed in the light of ethical considerations.

3. Ethical AI Design requires the confluence of many fields (technical, ethical, philosophical, legal, political)

Questions/Comments?

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